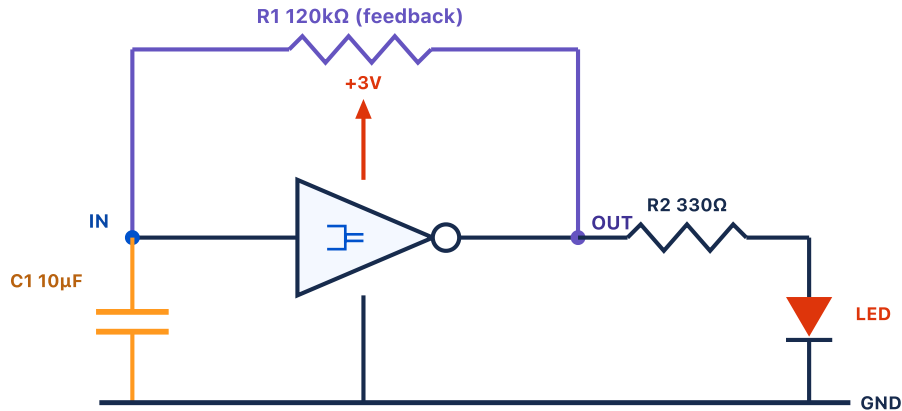


MagTile Circuits — Project 2: Be a Time Master

1-second LED blinker · single 74LVC1G14 Schmitt-trigger RC oscillator · 3 V CR2032 · printable build sheet

The circuit



Part 1 — Build the 1-second blink

1. Lay the **power rails**: tile at the left, +3 V on top, GND on the bottom.
2. Drop the **inverter** in the center — IN faces left, OUT faces right.
3. Run the **feedback** across the top: elbow → **R1 120 kΩ** → elbow back down to OUT. Put a **T** at IN and at OUT.
4. Hang **C1 10 µF** from the IN node down to GND.
5. From OUT, go through **R2 330 Ω** → elbow → **LED** → GND.
6. Power up. The LED should blink about **once per second** ("one-Mississippi").

Why it's one second

Quantity	Value
$\tau = R1 \cdot C1$	1.20 s
Schmitt factor k	0.83
Period $T = \tau \cdot k$	≈ 1.0 s
Blink rate	≈ 1 Hz

🔧 PART 2 — MAKE IT BLINK FASTER

Get it to **2 blinks per second**. Hint: the period is $\tau = R1 \times C1$ — which one tile do you shrink?

✅ ANSWER

Halve **either** tile: R1 120 kΩ → **62 kΩ**, or C1 10 µF → **4.7 µF**. Bigger = slower. R2 only sets brightness.

✂ Cut-out tiles — 1.5 in

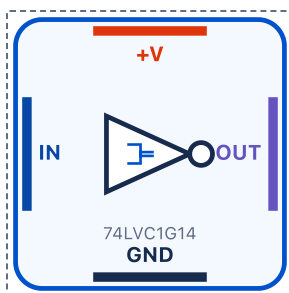
Each dashed square is exactly 1.5 in to match your tiles. Cut on the dashed lines and lay them out to match the schematic.

print-scale
check →

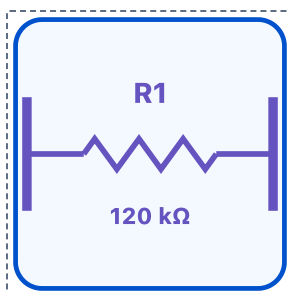
1 inch



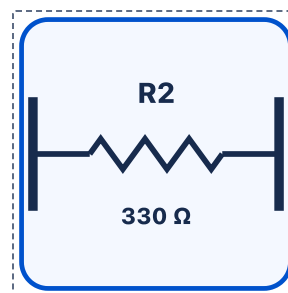
Power



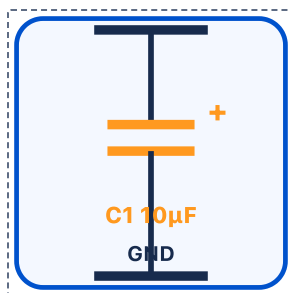
Inverter



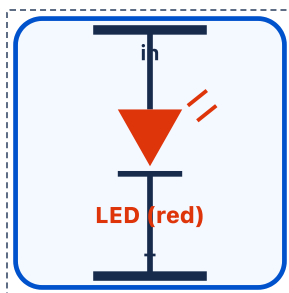
R1 feedback



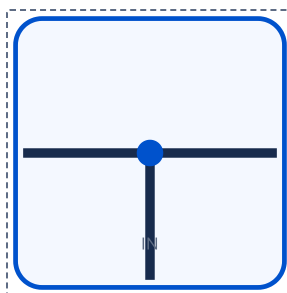
R2 LED limit



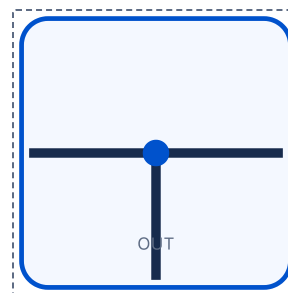
Capacitor



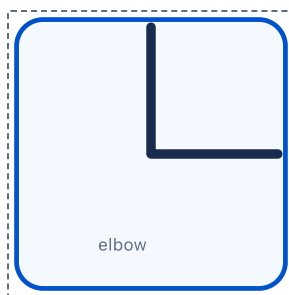
LED



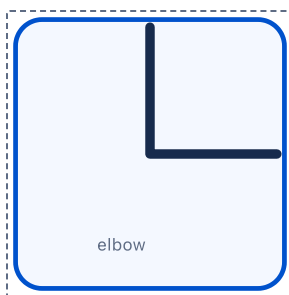
T — IN node



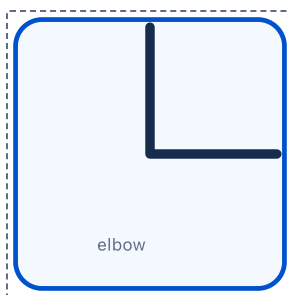
T — OUT node



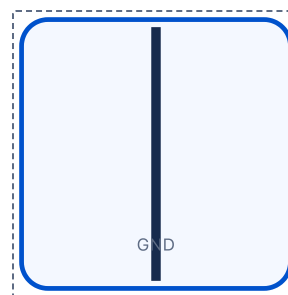
Elbow



Elbow



Elbow



Straight — GND

Print at **100% / Actual size** (no "fit to page"). Verify the 1-inch bar above measures 1 inch, then the tiles are true 1.5 in. · borenw.github.io/magtiles-circuits